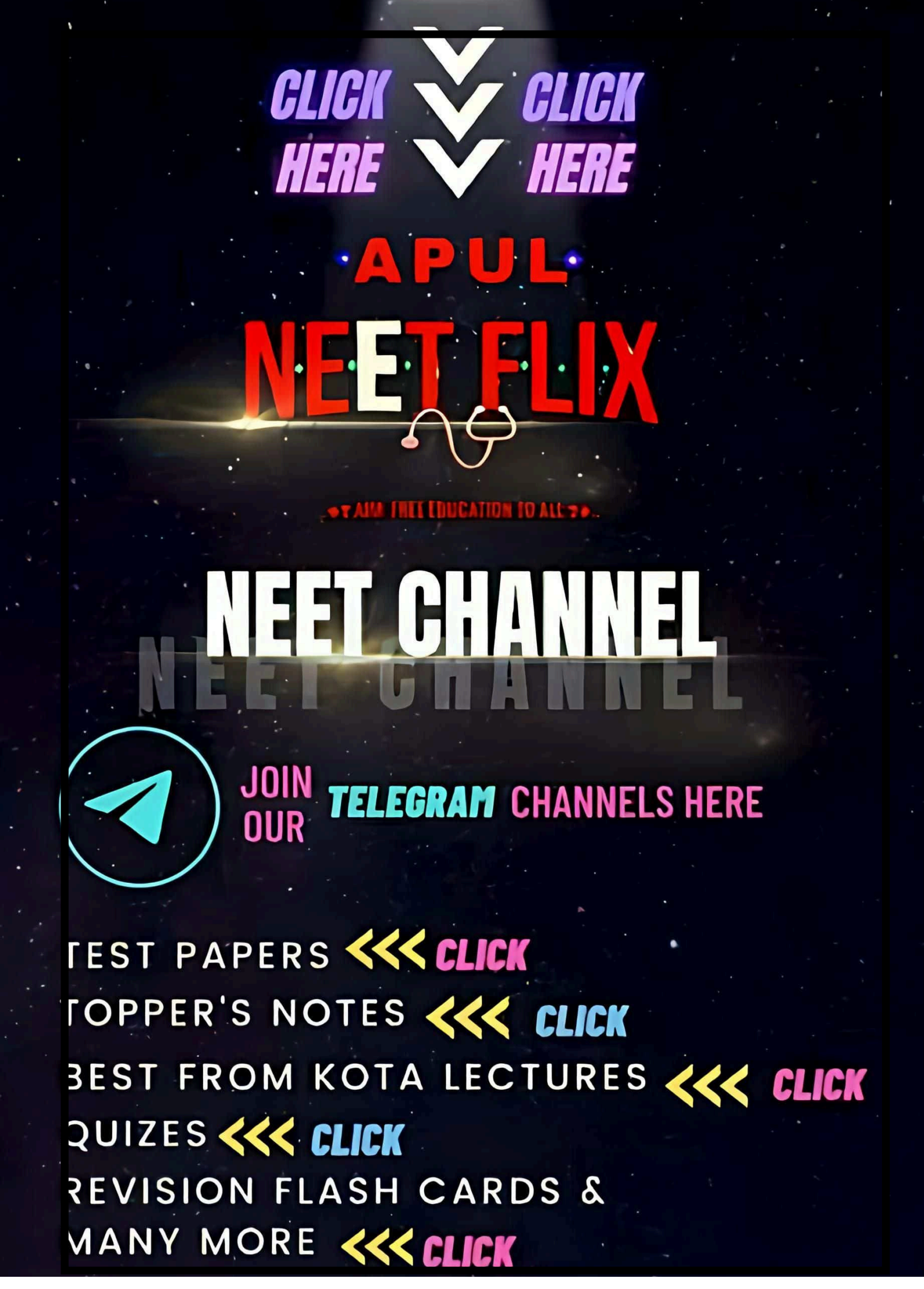

CLICK HERE
CLICK HERE
APUL
NEET FLIX
AIM FREE EDUCATION TO ALL
NEET CHANNEL
JOIN OUR TELEGRAM CHANNELS HERE
TEST PAPERS CLICK
TOPPER'S NOTES CLICK
BEST FROM KOTA LECTURES CLICK
QUIZES CLICK
REVISION FLASH CARDS &
MANY MORE CLICK

BIOLOGY

1) Which of the following are responsible for the production of methane in the gut of several ruminant animals :-

(1) Halophiles
(2) Acidophiles
(3) *E.coli*
(4) Methanogens

2) How many organisms are included in Monera in given list ?
(A) Slime moulds (B) Dinoflagellates
(C) Protozoans (D) Pseudomonas
(E) Archebacteria (F) Basidiomycetes

(1) 2
(2) 4
(3) 3
(4) 5

3) Which of the following character is similar in cyanobacteria and green plants :-

(1) Nitrogen fixation
(2) 80 s ribosme
(3) Chlorophyll 'a'
(4) Composition of cell wall

4) Single protective unit of bacteria is made up of:-

(1) Cell wall and cell membrane
(2) Glycocalyx, cell wall and flagella
(3) Cell wall, cell membrane and pili
(4) Capsule, cell wall and cell membrane

5) Bacterial structure and behaviour are respectively:

(1) Simple, Simple
(2) Complex, Simple
(3) Simple, Complex
(4) Complex, Complex

6) Which option is not related with mycoplasma ?

(1) Smallest living cell

(2) Faculative anaerobic

(3) Lack cell membrane

(4) Pathogenic in plants and animals

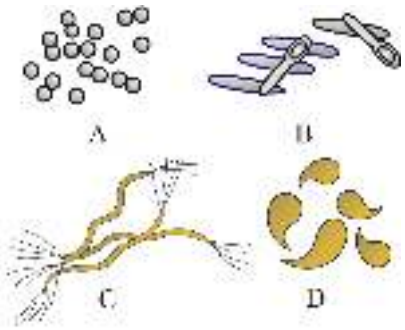
7) Choose the incorrect statement of following :-

(1) Ribosomes are the site of protein synthesis.

(2) Pili do not play a role in motility.

(3) Inclusion bodies are bounded by membrane.

(4) Cell wall prevent the bacterium from bursting and collapsing.

8) 

Select option showing correct identification for above figure from A to D :-

	A	**B**	**C**	**D**
(1)	*Bacillus*	*Coccus*	*Spirillum*	*Vibrio*
(2)	*Coccus*	*Bacillus*	*Spirillum*	*Vibrio*
(3)	*Coccus*	*Spirillum*	*Bacillus*	*Vibrio*
(4)	*Coccus*	*Vibrio*	*Bacillus*	*Spirillum*

(1) 1

(2) 2

(3) 3

(4) 4

9) Some characteristics are given below :
Unicellular, Aquatic, Autotrophic and heterotrophic mode of nutrition and asexual reproduction by binary fission, Eukaryotic
Above mentioned characters are related to which kingdom ?

(1) Monera

(2) Protista

(3) Fungi

(4) Animalia

10) Some characters are given below. Which of them are true for the cyanobacteria ?
(a) Oxygenic photosynthesis
(b) Non-oxygenic photosynthesis

(c) Oxygenic Chemotropism
(d) Ability of N_2 fixation

(1) a and b
(2) a and c
(3) a and d
(4) b and d

11) Archaebacteria differ from other eubacteria in having :-

(1) Both do not have true nucleus
(2) Composition of cell wall
(3) Composition of cytoplasm
(4) Composition of genetic material

12) Which of the following have capacity of N_2 fixation?

(1) *Anabaena*
(2) *Rhizobium*
(3) *Nostoc*
(4) All of these

13) Cell wall of bacteria is composed of :-

(1) Cellulose
(2) Glycogen
(3) Chitin
(4) Peptidoglycan

14) According to two kingdom system of classification Chlamydomonas and bacteria are placed under:-

(1) Kingdom – Protista, Monera
(2) Kingdom – Monera, Protista
(3) Kingdom – Fungi
(4) Kingdom – Plantae

15) Heterocyst is a structure which is associated with :-

(1) Reproduction
(2) Respiration
(3) Nitrogen fixation
(4) Locomotion

16) Which of the following fungus cause rust in wheat :-

(1) *Penicillium*
(2) *Puccinia*
(3) Mucor
(4) Yeast

17) Which of the following pigment is present in Euglenoid but lack in dinoflagellates and diatoms:-

(1) Chlorphyll 'a'
(2) Chlorophyll 'c'
(3) Chlorophyll 'b'
(4) Carotenes

18) Which of the following is anaerobic free living photosynthetic nitrogen fixing organism ?

(1) *Azotobacter*
(2) *Frankia*
(3) *Rhodospirillum*
(4) *Rhizobium*

19) According to R.H. whittaker, how many kingdoms are belong to eukaryotes ?

(1) Five
(2) Three
(3) Four
(4) Two

20) Which organism is responsible for red tide of sea ?

(1) *Red algae*
(2) *Gonyaulax*
(3) *Trichadesmium*
(4) *B.G.A.*

21) Which group show most extensive metabolic diversity :-

(1) Bacteria
(2) Bryophytes
(3) Algae
(4) Protista

22) Match the column A with column B and find out the correct answer :-

Column-A		**Column-B**	
(a)	Bacteria	(P)	Soap box like cell wall
(b)	BGA	(Q)	Mucilagenous sheath

(c)	Dinoflagellates	(R)	Plated cell wall
(d)	Chrysophyte	(S)	Plasmid

(1) a-S, b-R, c-P, d-Q

(2) a-S, b-R, c-Q, d-P

(3) a-Q, b-P, c-R, d-S

(4) a-S, b-Q, c-R, d-P

23) Which one of the following pairs is wrongly matched ?

(1) Fungi – Botulism

(2) *Archaebacteria* – Gobar gas

(3) *Saccharomyces* – Ethanol

(4) *Streptomycetes* – Antibiotic

24) Which of the following bacteria uses light energy for food synthesis ?

(1) *Chromatium*

(2) *Chlorobium*

(3) *Rhodopseudomonas*

(4) All of these

25) Some hyperthermophilic organisms that grow in highly acidic (pH2) habitats belong to the two groups:

(1) Liverworts and yeasts

(2) Eubacteria and archaea

(3) Cyanobacteria and diatoms

(4) Protists and mosses

26) The body of *Euglena* is flexible due to the presence of -

(1) Cell wall rich in polypeptide

(2) Protein rich layer called pellicle

(3) Cilia all around the cell envelope

(4) Protein less plasma membrane

27) In which of the following protists their overlapping shell fits together as in soap box?

(1) Diatoms

(2) Desmids

(3) Slime mould

(4) Sporozoans

28) Non-cellulosic cell wall is found

(1) Only in monerans

(2) Only in fungi

(3) Monerans & fungi

(4) Fungi and plants

29) In five kingdom classification, which of the following group of organisms is placed in the same kingdoms ?

(1) *Chlorella*, *Amoeba*, *Paramoecium*, *Spirogyra*

(2) *Chlamydomonas*, *Amoeba*, *Gonyaulax*, *Chlorella*

(3) *Spirogyra*, *Nostoc*, *Anabaena*, *Mycoplasma*

(4) *Spirogyra*, *Chlamydomonas*, *Euglena*, *Nostoc*

30) Which of the following kingdoms of five kingdoms classification system have heterotrophic organisms only ?

(1) Monera and Animalia

(2) Protista and Fungi

(3) Monera and Fungi

(4) Fungi and Animalia

31) Find out the incorrect statement–

(1) All the fungi are hetrotrophic

(2) Fungi prefer to grow in warm and humid places

(3) Wheat rust is cause by *Albugo*

(4) Yeast is used in making bread and beer

32) ***Assertion :-*** The cellwall of diatoms is indestructible.
Reason :- The cell wall of diatoms is embedded with calcium carbonates.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

33) **Assertion:** Archaebacteria Survive in extreme conditions.
Reason: Archaebacteria differ from other bacteria in having a different (complex) cell wall structure.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

34) **Assertion :** Spores in slime moulds are extremely resistant and survive for many years, even

under adverse conditions.
Reason : Spores do not possess true walls and true membranes.

(1) Both A and R correct and R is the correct explanation of A
(2) Both A and R are correct, but R is not correct explanation of A
(3) A is correct, but R is incorrect
(4) Both A and R are incorrect

35) Choose the **correct** options with respect to given diagram :



(1) Paramoecium → Protozoans, presence of thousand of cilia
(2) Gonyaulax → Dinoflagellates, one longitudinal flagella
(3) Nostoc → Cynobacteria, can fix N_2
(4) Euglena → Euglenoids. They have two flagella, a short and a long one.

36) Identify the organism given in the below figure





(1) *Euglena* and *Paramoecium*
(2) Slime mould and *Amoeba*
(3) Dinoflagellates and *Chlamydomonas*
(4) Dinoflagellates and Slime mould

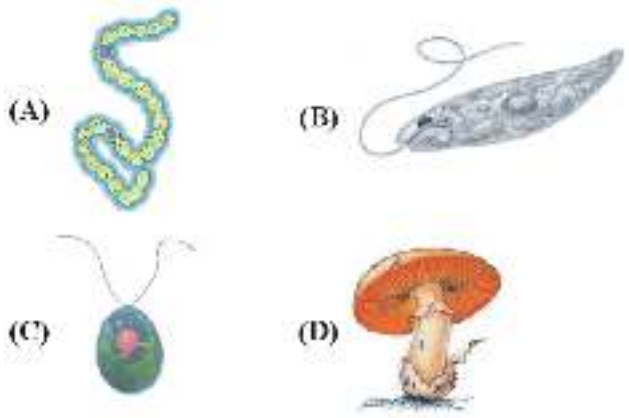
37)



Identify the organism and feature associated with it :-

(1) *Morchella* - Edible fungus
(2) *Agaricus* - Edible fungus
(3) *Rhizopus* - Bread mold
(4) *Alternaria* - Pathogenic fungus

38) Examine the figure (A-D) given below and select the right option in which all the four figure A, B,



C and D are identified correctly :

	A	B	C	D
(1)	*Nostoc*	*Euglena*	*Aspergillus*	*Spirullina*
(2)	*Nostoc*	*Euglena*	*Chlamydomonas*	*Basidiocarp*
(3)	*Nostoc*	*Paramoecium*	*Ascocarp*	*Chlamydomonas*
(4)	*Nostoc*	*Diatoms*	*Mucor*	*Chara*

(1) 1

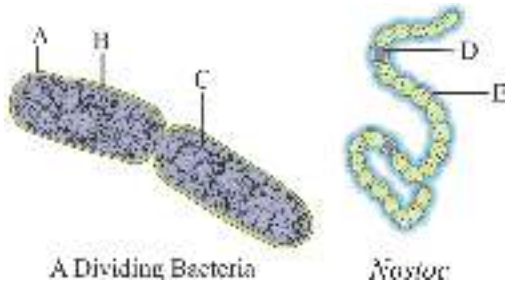
(2) 2

(3) 3

(4) 4

39)

Identify the A, B, C, D and E in the following figures.



(1) A-Cell wall, B-Cell membrane, C-Heterocyst, D-DNA, E-Mucilagenous sheath

(2) A-Cell wall, B-Cell membrane, C-DNA, D-Heterocyst, E-Mucilagenous sheath

(3) A-Mucilagenous sheath, B-Cell membrane, C-DNA, D-Heterocyst, E-Cell wall

(4) A-Cell membrane, B-Cell wall, C-DNA, D-Heterocyst, E-Mucilagenous sheath

40) Consider the following four statements (A-D) and select the option which includes all the **correct** ones only.

(A) Archaebacteria include methanogens halophiles and thermoacidophiles

(B) Bacteria as a group show the most extensive metabolic diversity
(C) Cyanobacteria have chlorophyll 'a' similar to green plants. (D) Mycoplasma can survive without oxygen.

(1) Statements B, C and D only
(2) Statements A, C only
(3) Statements B, D only
(4) Statements A,B,C and D

41) Choose the incorrect statement from following :-

(1) Bacteria show the most extensive metabolic diversity.
(2) Members of Protista are primarily aquatic
(3) Fungi show a great diversity in morphology and habitat.
(4) In five kingdom classification virus, viroids and prions are placed in kingdom monera

42) Choose the incorrect statement from following:-

(1) Yeast are used to make bread and bear
(2) Fungi prefer to grow in warm and humid places
(3) Cell wall of fungi are composed of chitin and polysaccharides
(4) Fungi show symbiotic association with algae as mycorrhiza and with roots of higher plants as lichens

43) How many of the following statements are true for fungi ?
(A) Members of kingdom fungi show great diversity in structure (morphology) and habitat
(B) Most of fungi are saprophytic in their mode of nutrition
(C) Asexual cycle in fungi involve three steps
(D) Morels and truffles are edible mushroom and are considered delicacies
(E) *Albugo* is the parasitic fungi on mustard and wheat
(F) Members of ascomycetes can be multicellular or unicellular

(1) Two
(2) Four
(3) Three
(4) Five

44) Choose the incorrect statement :

(1) Bacteria are the sole members of the kingdom Monera.
(2) Bacteria are grouped under four categories based on their shape.
(3) The bacterial structure is very simple, but are very complex in behaviour.
(4) The majority of bacteria are autotrophs.

45) Read the following statements and choose the correct option.
Statement-1 : Two kingdom classification system did not distinguish between the eukaryotes and

prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non photosynthetic (fungi) organsims.
Statement-2 : Cell wall of bacteria is made up of polysaccharide and amino acid and in fungi it is made up of chitin.

(1) Both statement-1 and 2 are correct
(2) Both statement-1 and 2 are incorrect
(3) Only statement 1 is correct
(4) Only statement 2 is correct

46) An isolated population with equal number of blue eyed and brown eyed individual was destroyed by earthquake, only a few brown eyed individual survived to form next generation. The change in gene pool is known as:

(1) Founder's effect
(2) Sexual selection
(3) Bottleneck effect
(4) Genetic equilibrium

47) Excretion is

(1) Removal of substances not required by body
(2) Removal of useless substances and susbtances present in excess
(3) Formation of substances having some role in body
(4) 1 and 2 both

48) Which of the following factors is responsible for the formation of concentrated urine?

(1) Low levels of antidiuretic hormone.
(2) Maintaining hyperosmolarity towards inner medullary interstitium in the kidneys.
(3) Secretion of erythropoietin by juxtaglomerular complex.
(4) Hydrostatic pressure during glomerular filtration.

49) Oxygen in atmosphere has been formed by :

(1) Evaporation of water
(2) Photosynthesis of blue green algae
(3) Metabolism of microorganisms
(4) Decaying organisms

50) A compound having very important role in prebiotic evolution was :

(1) SO_2
(2) CH_4
(3) SO_3
(4) NO

51) Due to discovery of which of the following in 1980, the evolution was termed as RNA world :

(1) m - RNA , t - RNA, r - RNA synthesise proteins
(2) In some virus RNA is genetic material
(3) RNA has enzymatic property
(4) RNA is not found in all cells

52) First life on earth was :-

(1) Cyanobacteria
(2) Chemoheterotrophs
(3) Autotrophs
(4) Photoautotrophs

53) Most modern hypothesis regarding origin of life was given by :

(1) Wallace
(2) Hugo de Vries
(3) Oparin
(4) Charles Darwin

54) Connecting link between annelida and mollusca :

(1) Cuttle fish
(2) *Octopus*
(3) *Neopilina*
(4) *Nautilus*

55) Homology is exhibited by :

(1) Wings of butterfly, birds and bat
(2) Paddle of whale, forearm of horse and forelimbs of man
(3) Tail of monkey and bird
(4) Sting of scorpion and honey bee

56) Which one of the following are analogous structures :-

(1) Wings of Bat and Wings of Pigeon.
(2) Gills of Prawn and Lungs of Man.
(3) Thorns of *Bougainvillea* and Tendrils of *Cucurbita*
(4) Flippers of Dolphin and Legs of Horse

57) Dinosaurs disappeared during :

(1) Jurassic
(2) Triassic

(3) Cretaceous
(4) Permian

58) Which of the following is closest to humans:-

(1) Gibbon
(2) Gorilla
(3) *Homo erectus*
(4) *Dryopithecus*

59) Which is the most common mechanism of genetic variation in the population of sexually reproducing organism?

(1) Chromosomal aberrations
(2) Genetic drift
(3) Recombination
(4) Transduction

60) S.L. Miller an American scientist, in his experiment, created electric discharge at 800°C in a closed flask containing some gases with water vapour. These gases were :

(1) Oxygen, Nitrogen, Hydrogen
(2) Methane, Ammonia, Nitrogen
(3) Methane, Ammonia, Hydrogen
(4) Oxygen, Methane, Ammonia

61) In Galapagos island, Darwin found different varieties of finches, originated from a common ancestor. The original variety of these finches was:-

(1) Insectivorous
(2) Vegetarian
(3) Seed-eating
(4) Cactus eating

62) Which statement is not true for protobionts?

(1) They were clusters of macromolecules
(2) They were isolated from their environment
(3) They could grow
(4) They could reproduce

63) Golden age of fish was :

(1) Devonian period
(2) Silurian period
(3) Triassic period

(4) Cambrian period

64) The process of evolution of different species in a given geographical area starting from a point and literally radiating to other area of geography (Habitat) is called :

(1) Convergent evolution
(2) Parallel evolution
(3) Adaptive radiation
(4) Both (1) and (2)

65) Biochemical evidance indicates towares :-

(1) Homology
(2) Divergent evolution
(3) Common ancestory
(4) All

66) According to Panspermia theory, life has :

(1) Come out of dead and decaying matter
(2) Originated from non-living organic molecules
(3) Come from another planet
(4) Created by God

67) The biggest dinosaur had about 20 feet height and huge, fearsome, dagger - like teeth. The name of it is

(1) *Ichthyosaurus*
(2) *Triceratops*
(3) *Tyrannosaurus*
(4) *Stegasaurus*

68) The chronological order of human evolution from first human like being-hominid to recent is:-

(1) *Homo habilis → Homo erectus → Neanderthal → Homo sapiens sapiens*
(2) *Australopithecus → Homo erectus → Homo habilis → Homo sapiens*
(3) *Homo erectus → Homo habilis → Neanderthal → Homo sapiens*
(4) *Australopithecus → Homo erectus → Homo sapiens → Ramapithecus*

69) About 200 mya, some reptiles moved back into water and evolved into :-

(1) Mammals
(2) Penguin
(3) Fish-like reptiles – *Ichthyosaurs*
(4) Amphibians

70) The different species evolved from common ancestors and living in different geographical regions are called

(1) sibling species
(2) sympatric species
(3) allopatric species
(4) polytypic species

71) Animal breeding and plant breeding programmes are the examples of :

(1) Co-evolution
(2) artificial selection
(3) mutation
(4) natural selection

72) What is main difference between the mutation of Hugo de Vries and variation of Darwin?

(1) According to Hugo de Vries mutations are random and directionless while Darwinian variations are small and directional.
(2) According to Hugo de Vries mutations are random and directional while Darwinian variations are small and directional.
(3) According to Hugo de Vries mutations are small and directional while according to Darwin variations are random.
(4) According to Hugo de Vries mutations are large and directional while Darwinian mutations are small, directionless.

73)

Which one is **correct** out of the following options ?

(1) The fossils of *Triceratops* was older than the fossils of *Stegosaurus*
(2) The fossils of *Stegosaurus* was older than the *Triceratops*
(3) The fossils of *Pteranodon* was older than the crocodilian
(4) The fossils of crocodilia was older than the fossils of *Archaeopteryx*.

74) In the developmental history of a mammalian heart, it observed that it is passed through a two chambered fish like heart stage, 3 chambered frog like heart stage & finally to four chambered heart stage. To which hypothesis can this above cited statement be approximated?

(1) Hardy - Weinberg law
(2) Lamarck's principle
(3) Biogenetic law
(4) Mendelian principle

75) Industrialization during 1920's in England led to

(1) Increase in population of white coloured biston moths.

(2) Increase in population of birds feeding on biston moths.

(3) Increase in population of dark coloured biston moths.

(4) Decrease in population of birds feeding on biston moths

76) Darwinism explains all the following except :

(1) offspring with better traits that overcome competition are best suited for the environment

(2) variations may or may not be inherited from parents to offspring through genes

(3) within each species, there are variations

(4) organisms tends to produce more number of offspring than can survive

77) Which one of the following scientist's name is correctly matched with the theory put forth by him?

(1) De Vries – Theory of natural selection

(2) Darwin – Mutation Theory

(3) Weismann–Theory of continuity of germplasm

(4) Pasteur–Theory of inheritance of acquired characters

78) Which of the following statement is correct ?

(1) Amphibians evolved into reptiles.

(2) Fish with stout and strong fins could move on land and go back to water. This was about 350 mya.

(3) Giant ferns were present but they all fell to form coal deposits slowly.

(4) All of these

79) Match the column-I with column-II and choose the correct option ?

Column-I		**Column-II**	
(A)	Weismann	(i)	Inheritance of acquired character
(B)	Hugo de Vries	(ii)	Biogenetic law
(C)	Ernst Haeckel	(iii)	Theory of continuity of germplasm
(D)	Lamarck	(iv)	Mutation theory

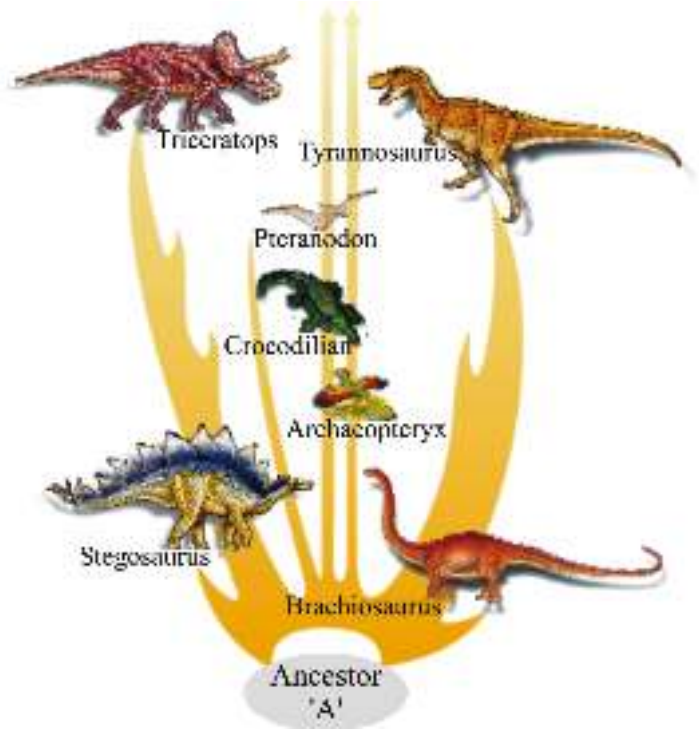
(1) A - iii B - i C - iv D - ii

(2) A - ii B - iii C - iv D - i

(3) A - iv B - ii C - i D - iii

(4) A - iii B - iv C - ii D - i

80) Identify the ancestor-A in the family tree of dinosaur in given figure :-



(1) Synapsids

(2) Thecodonts

(3) Therapsids

(4) Pelycosaurus

81) Which of the following amino acids were formed in Stanley-Miller's experiment ?

(1) Alanine, Arginine, Glutamic acid

(2) Arginine, Aspartic acid, Glutamine

(3) Aspartic acid, Alanine, Glycine

(4) Aspartic acid, Glutamic acid, Arginine

82) How many of the following statements are correct for gene migration or gene flow :-
(a) Gene frequencies change in the original population
(b) Gene frequencies change in the new population
(c) New gene/alleles are added to the new population
(d) Genes are lost from the old population

(1) Four

(2) Three

(3) Two

(4) One

83) Hardy-Weinberg equilibrium law states that the relative frequency of alleles in the population of sexually reproducing organisms remains constant from generation to generation if a. The population is large enough
b. random mating occurs
c. No mutation occurs

d. No migration occurs

(1) only a, b & c are correct
(2) only a, b & d are correct
(3) only a, b, c, d are correct
(4) only b, c & d are correct

84)

Which of the following is the correct sequence of events in the origin of life?
A. Formation of protobionts
B. Formation of macromolecules
C. Formation of DNA Based system

D. Formation of monomeric compounds
(1) A, B, C, D
(2) D, C, A, B
(3) D, B, A, C
(4) D, C, B, A

85) The cranial capacity of Peking man was about:

(1) 900 cc
(2) 1660 cc
(3) 1075 cc
(4) 1450 cc

86) Flippers of seal are modified :

(1) fins
(2) hindlimb
(3) forelimb
(4) gills

87) Theory of evolution is mainly concerned with :

(1) Spontaneous generation
(2) Theory of special creation
(3) Gradual change
(4) Conditions of environment

88) Prehistoric cave art developed about :

(1) 10000 years ago
(2) 15000 years ago
(3) 18000 years ago
(4) 20000 years ago

89) Branching descent is accounted for :

(1) Homology
(2) Analogy
(3) Vestigial organs
(4) Atavism

90) Alfred Wallace, a naturalist worked on which island ?

(1) Srilanka
(2) Galapagos
(3) South America
(4) Malay Archepelago

ANSWER KEYS

BIOLOGY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**4**	**1**	**3**	**4**	**3**	**3**	**3**	**2**	**2**	**3**	**2**	**4**	**4**	**4**	**3**	**2**	**3**	**3**	**3**	**2**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**1**	**4**	**1**	**4**	**2**	**2**	**1**	**3**	**2**	**4**	**3**	**3**	**1**	**3**	**4**	**4**	**2**	**2**	**2**	**4**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**4**	**4**	**3**	**4**	**1**	**3**	**2**	**2**	**2**	**2**	**3**	**2**	**3**	**3**	**2**	**2**	**3**	**3**	**3**	**3**
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	**3**	**4**	**1**	**3**	**4**	**3**	**3**	**1**	**3**	**3**	**2**	**1**	**2**	**3**	**3**	**2**	**3**	**4**	**4**	**2**
Q.	81	82	83	84	85	86	87	88	89	90										
A.	**3**	**1**	**3**	**3**	**1**	**3**	**3**	**3**	**1**	**4**										

SOLUTIONS

BIOLOGY

1)

Explain Question : The question asks to identify the specific type of microorganisms responsible for producing methane gas in the digestive system (gut) of ruminant animals.

Concept : This question is based on Archaebacteria.

Solution : "Methanogens are a group of archaebacteria that produce methane in the guts of ruminants."

Final Answer : option (4)

2)

Solution :

The Monera kingdom includes prokaryotic organisms, such as:

A. **Pseudomonas (a bacterium)**

B. **Archaebacteria (a type of archaea)**
Answer :- 1

3)

Explain Question : The question asks to identify a characteristic that is shared between blue-green algae (cyanobacteria) and green plants.

Concept : This question is based on cyanobacteria.

Solution : "Both blue-green algae (cyanobacteria) and green plants contain chlorophyll 'a'."

Final Answer : option (3)

4) NCERT (XI) Pg. # 128

5) NCERT (XI) Pg. # 19

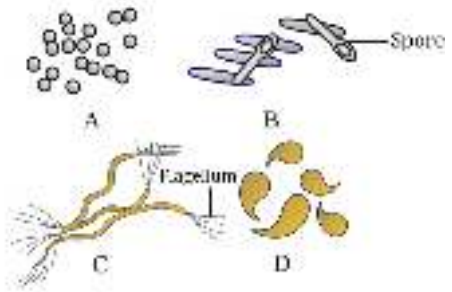
6) NCERT XI Pg. # 20

7)

NCERT-XI, Pg. # 128, 129

8)

Explain Question :
The question is asking you to identify the shapes of bacteria labeled as A, B, C and D in a diagram and match them with the correct bacterial forms from the given options.



Concept :
This question is based on Bacteria of different shapes.

Solution :
The correct answer is option 2- Coccus, Bacillus, Spirillum, Vibrio
The image in the diagram show:
A: Round → Coccus
B: Rod-shaped → Bacillus
C: Spiral → Spirillum
D: Comma-shaped → Vibrio

Final Answer : option (2) – Coccus, Bacillus, Spirillum, Vibrio

9)

NCERT-XI, Pg. # 15

10)

SOLUTION—

While some cyanobacteria can perform anoxygenic photosynthesis, it is not a universal characteristic. Oxygenic chemotropism is not a feature of cyanobacteria; chemotrophs obtain energy from chemical compounds, not light.

The correct answer is: OPTION(3) a and d

11) NCERT (XI) Pg # 19

12)

SOLUTION—

Anabaena, Rhizobium, and Nostoc all have the capacity for nitrogen fixation.

The correct answer is: OPTION(4) All of these

13)

NCERT XI, Pg # 17, Table 2.1

14) NCERT XI, Pg # 16

15)

NCERT-XI, Pg. # 19, Eubacteria section

16) NCERT Pg. # 22

17)

NCERT XI Page No. # 21

18)

Module No. 1 Pg # 15

19)

According to R.H Whittaker's five-kingdom classification system, four kingdoms belong to eukaryotes: Protista, Fungi, Plantae, and Animalia.

Correct Answer is option (3)

20) NCERT (XIth) Pg # 21

21) NCERT (XI) Pg. # 19

22)

NCERT XI pg.# 19, 20, 21

23)

Module No. 1 Pg # 20

24)

Module No. 1 Pg # 22

25)

Question Explanation :
Which hyperthermophilic organisms grow in highly acidic (pH_2) habitats.

Concept :
This question is based on Such organisms belongs to eubacteria and archaea.

Solution :
Eubacteria and Archaea. These organisms are commonly found in hot springs, hydrothermal vents, and acidic volcanic environments. Eubacteria (e.g., Thermoplasma, Acidithiobacillus) Archaea (e.g., Sulfolobus, Pyrococcus, Thermococcus)

Hence,
Option (2) is correct : Eubacteria and archaea.

26)

NCERT Pg # 21

27)

NCERT Pg # 25

28)

NCERT Pg # 18

29)

NCERT-XI, Pg # 17, 21

30)

NCERT Pg # 25

31) **Explanation :-** The question presents four statements about fungi, and the task is to identify the incorrect one.
Concept :- This question is based on Characteristics and classifications of fungi
Solution :- Wheat rust is caused by Albugo - Incorrect. Wheat rust is caused by fungi of the genus Puccinia, not Albugo. Albugo species cause white rust diseases, primarily affecting cruciferous plants like cabbage and mustard, not wheat.

Answer :- 3

32)

Solution:

The correct answer is 3. Assertion is true but Reason is false.

A. Assertion: True (the cell wall of diatoms, called a "frustule," is made of silica, which is very durable and nearly indestructible).

Reason: False (the cell wall of diatoms is made of silica, not calcium carbonate).

33) NCERT Pg. # 19

34) NCERT Page. 21

35) NCERT Pg. # 21 (Para-2)

36)

NCERT-XI, Pg # 21

37) **Explain Question:** match the organism with its correct feature or description.
Concept: This question is based on basidiomycetes
Explanation: Agaricus - Includes common edible mushrooms. Edible fungus
Final answer: Agaricus - Edible fungus

38) NCERT XI Pg # 19, 21, 23, 31 Fig. 2.2, 2.4(a), 2.5(c), 3.1(a-ii)

39)

NCERT XI Pg. # 13,14

40)

NCERT XI Page No. # 13

41) NCERT-XI Pg. # 19,20,22,25

42)

NCERT XI, Pg. # 22, 23

43) A, B and F are correct.
NCERT XI Pg. # 23, 24 and 28

44) NCERT-XI, Pg. # 18 & 19, [2.1]

45) NCERT Pg#16,17

46)

Earthquake reduces population size drastically → Bottleneck effect.

47) **Question is Asking About:**
Excretion is

Solution/Explanation:
Removal of metabolic waste (useless). Substance and Substances prevent in excess.

48)

NCERT PAGE NO. 211

49)

Oxygen in atmosphere came from photosynthesis by cyanobacteria.

50)

Prebiotic evolution important molecule (Methane).

51)

RNA World concept due to enzymatic property of RNA.

52)

First life forms were chemoheterotrophs.

53)

Modern origin of life hypothesis - Oparin.

54)

Connecting link between annelids & molluscs= Neopilina.

55)

Homologous organs = whale flipper, horse forelimb, human arm.

56)

Analogous organs = thorns of Bougainvillea & tendrils of Cucurbita.

57)

Dinosaurs disappeared in Cretaceous period

58)

Closest to humans= Gorilla.

59)

Genetic variation main cause = Recombination.

60) NCERT(XII) Pg# 127 Para: 7.1
Miller used gases = CH_4, NH_3, H

61)

Original Darwin's finches were seed-eaters.

62)

Protobionts could grow but not reproduce.

63)

Golden age of fishes = Devonian period.

64)

Evolution radiating in new habitats = Adaptive radiation.

65) NCERT (XII) Pg. # 131
Biochemical evidence - Common ancestry (all correct).

66) NCERT Pg. No. 127
Panspermia: life came from outer space.

67) **Explaining the Question:**
Type of dinosaur
Concept:
Brief account of evolution
Solution:
The Biggest Dinosaur had about 20 feet height and huge, fearsome, dagger like teeth. The name of it is Tyrannosaurus.
Final Answer:
Option (3) Tyrannosaurus **OR** Biggest dinosaur with dagger teeth Tyrannosaurus.

68) NCERT (XII), Pg # 140, 141
Human evolution correct order = H. habilis - H. erectus - Neanderthal - H. sapiens sapiens

69)

Some of the land reptiles went back into water to evolve into fish like reptiles probably 200 mya (e.g. Ichthyosaurs).

OR Reptiles re-entered water ~200 mya → Ichthyosaurs.

70)

NCERT-XII, Pg. # 121
Different species, different geography, same ancestor = Allopatric species.

71)

The correct answer is **2. artificial selection.**

A. Artificial selection is the process by which humans selectively breed organisms with desirable traits. This is done by choosing individuals with specific characteristics to be parents of the next generation.

OR Animal & plant breeding - Artificial selection.

72) According to Hugo de Vries it is mutation which causes evolution and not the minor variations (heritable) that Darwin talked about.

A. Mutations are random and directionless while Darwinian variations are small and directional.

OR Difference: De Vries mutations = sudden, random, large; Darwin's variations = small, directional.

73)

NCERT (XII) Page no. 114
Fossil older - Stegosaurus older than Triceratops.

74)

- The Biogenetic law, also known as Recapitulation theory, was proposed by Ernst Haeckel.

- It suggests that the developmental stages of an embryo (ontogeny) reflect the evolutionary history (phylogeny) of the species.
- In this context, the observation of the mammalian heart passing through stages resembling a fish-like two-chambered heart, a frog-like three-chambered heart, and finally a four-chambered heart in mammals aligns with this hypothesis.

The correct answer is option 3. Biogenetic law.

OR Mammalian heart stages support Biogenetic law.

75)

Explain Question : to identify the one that is caused by Industrialization during 1920's in England

Concept : example of natural selection

Explanation :Natural selection favored the dark-colored moths, leading to a rise in their population during industrialization.

Final answer :3

OR

Industrialization increased dark moths population

76)

Darwinism doesn't explain inheritance mechanism
(variations not always genetic).

77)

NCERT : PAGE 135
Correct match = Weismann - continuity of germ plasm.

78) The correct answer is **4. All of these**
1. Amphibians evolved into reptiles.

A. This is true! Amphibians were the first vertebrates to colonize land, and over time, some lineages evolved adaptations that allowed them to live fully on land, giving rise to the reptile group.

2. Fish with stout and strong fins could move on land and go back to water. This was about 350 mya.

A. This also describes a key event in vertebrate evolution. Around 350 million years ago, certain lobe-finned fish developed strong, fleshy fins that enabled them to move on land and return to water. This was a crucial step in the transition from water to land.

3. Giant ferns were present but they all fell to form coal deposits slowly.

A. This is accurate. During the Carboniferous period, large fern-like plants flourished in swampy environments. When these plants died and were buried under layers of sediment, they underwent a process called coalification, eventually forming the coal deposits we mine today.

OR Correct statements: all (fish to land, giant ferns, amphibians - reptiles).

79)

The correct answer is indeed Option 4:
• A-iii: Weismann proposed the Theory of Continuity of Germ plasm.
• B-iv: Hugo de Vries is known for his Mutation Theory.
• C-ii: Ernst Haeckel proposed the Biogenetic Law.
• D-i: Lamarck is famous for his theory of Inheritance of Acquired Characteristics.

OR Match: A-iii, B-iv, C-ii, D-i.

80) NCERT XII Pg. # 114
Dinosaur ancestor= Thecodonts.

81)

NCERT (XII) Page no. 112
Allen module

OR Miller experiment formed Aspartic acid, Alanine, Glycine.

82)

Gene flow effects - all four correct.

83)

Hardy-Weinberg equilibrium requires large pop., random mating, no mutation, no migration - all.

84)

Origin of life events: Monomers → Macromolecules → Protobionts → DNA system.

85)

Peking man cranial capacity ≈ 900 cc.

86)

Seal flippers are modified forelimbs.

87)

Evolution theory mainly about gradual change.

88)

Cave art developed~ 18,000 years ago.

89)

Branching descent shown by homology.

90)

NCERT : PAGE 129
Alfred Wallace worked in Malay Archipelago.